Farah Aymen

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Cover Statement

I am currently an Assistant Lecturer at the Faculty of Informatics and Computer Science at the British University in Egypt. I hold an M.Sc. in Artificial Intelligence with First Class Honors (GPA 4.0, Valedictorian), and I have an active publication record, including a Best Paper Award (FICTA 2025). My research has developed around representation learning and the mathematical foundations of machine learning, with strengths in deep learning, NLP, and LLMs, multimodal systems, and reinforcement learning. I have experience in designing and analyzing learning systems, with a focus on generalization, interpretability, and model behavior across structured and sequential data. I am seeking to pursue a PhD to further develop these research directions and contribute to the development of robust, interpretable, and adaptive intelligent systems.

Education

M.Sc. Computer Science (Artificial Intelligence) — Sept 2023 — Dec 2025
London South Bank University (UK) & The British University in Egypt

- First Class with Honors, **GPA 4.0/4.0** — **Valedictorian**
- Thesis: *Enhancing Kolmogorov–Arnold Networks with Alternative Polynomial Classes: A Comparative Analysis Against Advanced Deep Learning Techniques*

B.Sc. Computer Science (Artificial Intelligence Major) — Sept 2019 — May 2023
London South Bank University (UK) & The British University in Egypt

- First Class with Honors, **GPA 4.0/4.0**

Research Experience

Kolmogorov–Arnold Networks for ECG Forecasting and PDE Solving — June 2024 — May 2025

- Replaced B-spline parameterizations with orthogonal polynomial bases (**Chebyshev, Hermite, Legendre**) to study approximation stability and expressivity.
- Designed KAN-based architectures for ECG forecasting (**MIT–BIH dataset**) and neural solvers for **Burgers' equation**.
- Achieved comparable predictive accuracy with **50% fewer parameters**; evaluated via MSE, MACs, and runtime benchmarks.
- Built reproducible training/evaluation pipelines with experiment tracking and controlled seeds.

Protein Graph Classification using Graph Convolutional Networks (GCN) — Feb 2024 — May 2024

- Developed a GCN pipeline using **PyTorch Geometric** for protein structure classification.
- Implemented clique-detection and graph feature extraction to characterize structural subgraphs.
- Produced an interpretable analysis linking graph topology to classification outcomes.

Publications

Authored: ***5*** | *Total impact:* ***1,700+ reads | 30+ citations***

- F. Aymen, A. Pester, M. Muttardi (2025). ***"KANs: Comparative Analysis of Polynomial Bases for ECG Time-Series Forecasting."*** FICTA, Manchester, UK. **Best Paper Award**. — **5 reads**
- F. Aymen, W. Hussein (2024). ***"Application of Spatial and Wavelet Transforms for Improved Deep Fake Detection."*** AIRC, pp. 13–17. — **34 reads | 5 citations**
- F. Aymen, H. Monir, A. Pester (2024). ***"Large Vision Models: Transformer-based Models vs Traditional Architectures in Video Processing."*** AIRC, pp. 50–54. — **38 reads | 2 citations**
- A. Abo-El-Rejal, S. E. Ayman, F. Aymen (2024). ***"Advances in Breast Cancer Segmentation: A Review."*** Acadlore Trans. AI & ML, vol. 3, no. 2, pp. 70–83. — **1,546 reads | 35 citations**
- F. Aymen, A. Pester, F. Andres (2023). ***"Synthetic Water Crystal Image Generation Using VAE-GANs and Diffusion Models."*** LNNS, vol. 936, pp. 95–104. — **76 reads | 3 citations**

Awards & Distinctions

- Best Paper Award — FICTA 2025
- Valedictorian — M.Sc. Computer Science (AI)
- IELTS – Score: 7.5
- ECPC Finalist — Egyptian Collegiate Programming Contest (2x)

Teaching Experience

Assistant Lecturer, The British University in Egypt — Cairo, Egypt
Feb 2026 — Present

- Promoted from Teaching Assistant in recognition of completing the Master's Degree.
- Delivered lectures and advanced labs in NLP, AI Planning, and Logic & AI.
- Supervised capstone projects and assessed coursework and research presentations.

Teaching Assistant, The British University in Egypt — Sept 2023 — Jan 2026

- **Natural Language Processing (80+ students):** Delivered weekly labs on tokenization, embeddings, attention, Transformers, and retrieval-augmented generation, and Agentic systems, and gave weekly support to deployable projects.
- **AI Planning for Robotic Systems (80+ students):** Taught Q-learning and Deep Q-learning pipelines; supported experimental evaluation.
- **Operating Systems (200+ students):** Delivered Linux labs on scheduling, synchronization, and concurrency.
- **Logic and AI (100+ students):** Taught search algorithms and metaheuristics; supervised project demos.

Academic Service

- Designed and delivered technical and non-technical workshops and research talks for academic and public audiences.
- Course Instructor, School of Continuing Education (BUE Chapter): Foundations of AI; Applications of AI.

Technical Expertise

- **Neural Architecture Design & Scientific ML:** Structured neural models, approximation-theoretic analysis, physics-informed learning, neural PDE solvers, stability–expressivity evaluation.
- **Deep Learning & Generative Modeling:** PyTorch (primary), TensorFlow/Keras; transformers, diffusion models, multimodal and representation learning.
- **Computer Vision:** CNNs, Vision Transformers, feature extraction, generative image modeling, deepfake detection pipelines.
- **Natural Language Processing:** Embeddings, attention mechanisms, transformer architectures, retrieval-augmented generation (RAG), language model experimentation.
- **Reinforcement Learning:** Q-learning, Deep Q-learning (DQN), value-function methods, policy-gradient methods, stochastic approximation, Actor-Critic.
- **Data Science & Statistical Modeling:** Exploratory data analysis, feature engineering, time-series modeling, statistical evaluation, and benchmarking.
- **Graph & Knowledge-Based Systems:** Graph neural networks (GCNs), knowledge graphs (Neo4j), graph-based retrieval pipelines.
- **Programming & Infrastructure:** Python, C/C++, MATLAB, CUDA; Linux, Git, Docker; reproducible ML experimentation workflows.

Certifications

Google Data Analytics Professional Certificate — 2022
Coursera

Mathematics for Machine Learning — 2024
Imperial College London (Coursera)

Referees

Available upon request.